

Student Handout

■ Final Exam – Applications of Large Language Models

****Weight:**** 20% • ****Duration:**** 3 hours • ****Format:**** Programming (open book/internet; any LLM/agent allowed)

****Submission:**** `.ipynb`, `.pdf`, and two CSVs (Part A & Part B)

Environment

A Colab notebook is provided with the baseline RAG (FAISS + SentenceTransformers + FLAN-T5). No setup required.

Dataset

****Title:**** *History and Impact of Renewable Energy Technologies* (~2 pages)

Stored at `exam/dataset/renewable\_energy\_exam.txt`

Exam Queries (5)

- 1) Country + year of first modern wind turbine
- 2) Two environmental challenges in the text
- 3) How to mitigate wind variability outages (from “Future Trends”)
- 4) Summarize pros/cons of solar, wind, hydro (3–4 sentences)
- 5) Which tech has highest social/environmental trade-offs and why

Part A — Prompting & Decoding (50)

- Make ****two prompt changes**** (persona, format, reasoning).
- Try ****two decoding strategies**** (temperature/topk/topp).
- Compare baseline vs. modified on 5 queries; fill the ****Part A table**** (1–4 Correctness & Coherence).
- Write a ****3–4 sentence reflection**** on tool use.

Part B — Retrieval & Evaluation (50)

- Choose ONE: change ****chunking****, swap ****embedding model****, or add ****reranking****.
- Evaluate ****Precision/Recall**** (context) and ****Answer Relevance (1–4)****; fill the ****Part B table****.
- Write a ****3–4 sentence reflection**** on tool use.

Optional Bonus (+10) – Agentic Twist

Choose one: Self-Ask, Plan and Answer, or Tool Selector rule; run on one query; add a short note.

Scoring (per Part = 50)

- Code 20 • Commentary 10 • Evaluation 10 • Reflection 10
- **Total:**** 100 (+10 bonus possible)

Submission Checklist

- [] Part A done + table filled
- [] Part B done + table filled
- [] Reflections written
- [] (Optional) Bonus attempted
- [] Exported `.ipynb`, `.pdf`, and both CSVs